RAG Learning

Required Practicals — OCR B GCSE Physics

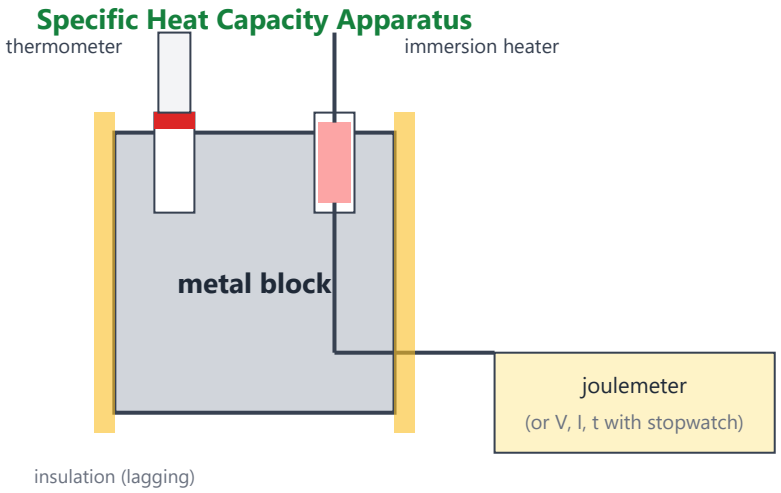
OCR GCSE Physics B (J259) — Apparatus, techniques, methods, analysis. Questions on *any* practical assessed across both papers.

The OCR GCSE specification requires schools to cover a core set of practical activities. Exam papers include questions on these — you'll need to describe methods, identify independent/dependent/control variables, evaluate precision/accuracy/repeatability, and analyse results. This page summarises the eight most examined practicals.

Practical 1 — Specific Heat Capacity (P2/P6)

Aim: Determine the specific heat capacity of a metal block.

- **Independent variable**: time / energy supplied.
- **Dependent variable**: temperature of the block.
- **Controls**: mass of block, starting temperature, amount of insulation.



Method:

1. Measure the mass **m** of the block using a top-pan balance.
2. Insert the immersion heater and thermometer into the holes. Add a drop of oil to the thermometer hole for good thermal contact.
3. Record the initial temperature θ_1.
4. Connect the heater to the joulemeter (or in series with an ammeter and voltmeter). Switch on and start a stopwatch.
5. Every minute, record temperature for 10 minutes.
6. Record final temperature θ_2 and energy input **E** from the joulemeter (or calculate E = V × I × t).

$$c = E / (m \times \Delta\theta)$$

Safety: Wear eye protection. The heater and block become hot — do not touch. Let cool before removing.

Practical 2 — Density of a Solid and a Liquid (P6)

Aim: Measure the density of a regular solid, an irregular solid and a liquid.

Method: Regular solid (e.g. cuboid):

1. Measure mass with a balance (g).
2. Use a ruler or vernier callipers to measure length, width, height (cm).
3. Volume = l × w × h. Density = m / V.

Irregular solid (displacement can):

1. Fill an eureka can with water to the spout.
2. Place a measuring cylinder below the spout.
3. Gently lower the object in; catch displaced water. Volume = volume collected.

Liquid:

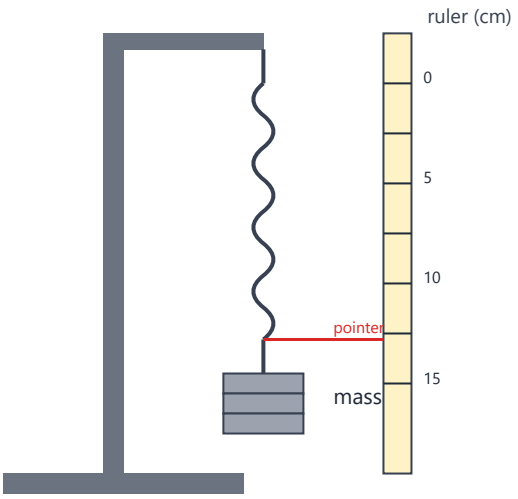
1. Place an empty measuring cylinder on the balance and zero it.
2. Pour liquid in and read volume and mass simultaneously. Density = m / V.

Exam Tip: Measure volumes in cm^3 and masses in g, then density is in g/cm^3. Multiply by 1000 for kg/m^3.

Practical 3 — Forces and Extension (Hooke's Law) (P4/P6)

Aim: Investigate the relationship between force applied to a spring and its extension.

Hooke's Law Apparatus



Method:

1. Hang a spring from a clamp stand. Attach a pointer to the bottom of the spring.
2. Measure the unloaded position l_0 with the ruler next to the pointer.
3. Add a 100 g mass (≈ 1 N) to the hanger. Record the new pointer position l.
4. Add masses one at a time up to at least 6 (6 N). After each addition, record position.
5. Extension $x = l - l_0$ for each force.
6. Plot a graph of force (y) against extension (x). A straight line through origin confirms Hooke's Law. Gradient = spring constant k.

Safety: Take care — if the spring breaks it can recoil. Place a "crash pad" under the hanger in case masses fall.

Practical 4 — Acceleration (P4)

Aim: Investigate how force and mass affect the acceleration of a trolley.

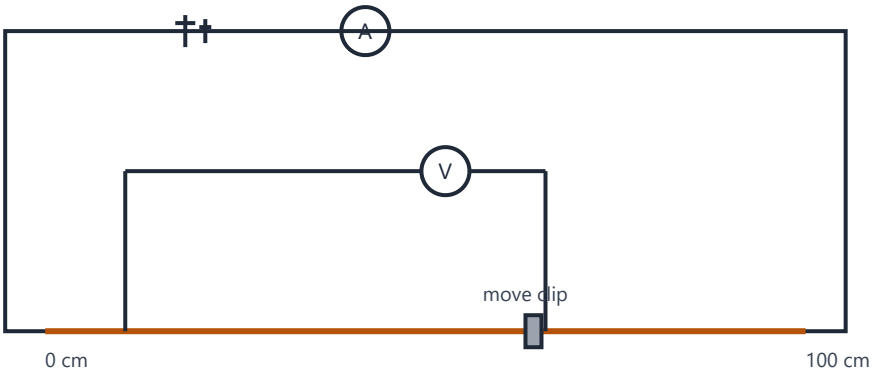
Method: Using light gates and a computer / data logger:

1. Set up a ramp with a trolley. Attach an interrupt card of known length to the trolley.
2. Place two light gates a fixed distance apart along the trolley's path.
3. Use a pulley system with a known weight pulling the trolley (this is the accelerating force).
4. Release the trolley. The light gates and timer record the time at each gate, giving speed before (u) and after (v) and the time between (t).
5. Calculate $a = (v - u) / t$.
6. **Vary the force** by moving masses from the trolley to the hanger (keeping total mass the same — total m constant).
7. **Vary the mass** by adding masses to the trolley while keeping the accelerating force constant.
8. Plot a against F (expect straight line through origin, confirming $F = ma$) and a against $1/m$ (expect straight line).

Practical 5 — Resistance of a Wire (P3)

Aim: Investigate how the resistance of a wire depends on its length.

Resistance of a Wire — Circuit



Method:

1. Set up the circuit with a battery, ammeter in series, and a 1 m length of constantan (or nichrome) wire taped to a metre rule.
2. Connect a crocodile clip at 10 cm. Place the voltmeter across the length of wire between the fixed end and the crocodile clip.
3. Close the switch briefly, record V and I, and open the switch again (to avoid the wire heating up).
4. Calculate R = V / I for this length.
5. Repeat for at least 6 different lengths (20, 30, …, 100 cm).
6. Plot R against L. A straight line through the origin confirms R ∝ L.

Exam Tip: Keep the current low (< 0.5 A) and turn off between readings — otherwise the wire heats up and resistance rises, giving systematic error.

Practical 6 — I–V Characteristics (P3)

Aim: Investigate the I–V characteristics of a filament lamp, a fixed resistor and a diode.

Method:

1. Set up a circuit with the component in series with a variable resistor, ammeter and cell. Voltmeter in **parallel** with the component.

2. Adjust the variable resistor to give a range of p.d. values across the component. Record V and I at each step.
3. Reverse the cell (swap connections to battery) to obtain negative values.
4. Plot I (y) against V (x). Expected shapes:
   — Fixed resistor: straight line through origin (ohmic).
   — Filament lamp: S-shape through origin (R rises with temperature).
   — Diode: zero below ~0.7 V, then sharp rise. No current in reverse.

Practical 7 — Infrared Radiation (Leslie's Cube) (P2)

Aim: Investigate the emission of IR radiation from different surfaces.

Method:

1. Fill a Leslie's cube (a hollow metal cube with four different surfaces: matt black, matt white, shiny silver, shiny black) with boiling water.
2. Place an IR detector (or thermopile) a fixed, short distance from each face in turn.
3. Record the IR reading from each face.
4. Keep: distance from cube constant, water temperature constant, surrounding temperature and conditions the same.

Expected results: matt black emits the most IR; shiny silver emits the least. Conversely, matt black is the best *absorber*, shiny silver the best *reflector*. This is why solar panels are matt black and space-suits are silvered.

Practical 8 — Waves in a Ripple Tank / on a String (P1)

Aim: Measure the frequency, wavelength and speed of waves in water and on a stretched string.

Method: Ripple tank (water waves):

1. A motor with an eccentric wheel creates a continuous wave pattern on the water surface. A stroboscope or slow-motion camera freezes the pattern.
2. Measure the distance across several wavelengths and divide — reduces uncertainty. Record λ.
3. Count the number of waves passing a point in 10 s → f.
4. v = f × λ.

Stretched string (microphone + signal generator loudspeaker):

1. A signal generator drives a loudspeaker with a known frequency f. A vibrating string of length L resonates at its fundamental.
2. Wavelength of the fundamental = 2L. Measure L with a ruler.
3. v = f × λ.

Practical 9 — Half-life Analogue (P5)

Aim: Model radioactive decay using dice or coins.

Method:

1. Start with 100 dice in a beaker. Each die = one "atom".
2. Roll all the dice onto the table.
3. Remove the dice that show a 6 — these have "decayed".
4. Count the remaining dice, record, and repeat.
5. Plot number remaining (y) against throws (x). A smooth exponential curve. Use the graph to read off how many throws it takes for the count to fall to 50, 25, 12.5 — these intervals are equal, confirming constant half-life.

This analogy illustrates that decay is **random** but **statistically predictable** — exactly the pattern observed in real radioactive samples.

Practical Skills — Everything You Need to Know

Skill	What examiners expect
Variables	Identify independent, dependent and control variables
Accuracy	How close to the true value (use calibrated instruments, correct zero errors)
Precision	Spread of repeated readings (small spread = precise)
Repeatability	Same person, same equipment → similar results
Reproducibility	Different person/equipment → similar results
Resolution	Smallest change a measuring instrument can detect
Anomalies	Readings that don't fit the pattern — investigate and possibly exclude from the mean
Uncertainty	± half the resolution for analogue; ± the last digit for digital (rule of thumb)

Exam Tip: **Common mark-scheme hooks:** "measure with a *suitable* instrument" (e.g. micrometer for wire diameter), "repeat and take a mean", "plot a graph to identify a relationship", "identify a control variable", "evaluate the limitations of the method".

© RAG Learning · Not for redistribution